

# Expected Linear Time Algorithm for Computing 3D Relative Neighborhood Graphs

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## ABSTRACT

An expected linear time algorithm for computing the relative neighborhood graph (RNG) for points in Euclidean space, distributed uniformly in a unit three dimensional cube, is given. For each point, potential neighbors for an RNG connection are limited to an expected small bounding volume. Volumes relative to the considered point are bounded through half plane partitions, constructed from the point and its neighbors, that remove regions where no RNG edge connections to the point can occur. The small bounding volume provides an expected constant number of neighbors to consider, making each single point RNG calculation  $O(1)$ , providing the  $O(n)$  expected runtime when enumerating through all points. To the author's knowledge, this improves on the previously best known  $O(n \log n)$  in three dimensions.

## 1. Introduction

Let  $P = (p_0, p_1, \dots, p_{n-1})$  be a set of points in  $\mathbb{R}^3$ . The relative neighborhood graph,  $\text{RNG}(P)$ , is an undirected, simple graph that connects two vertices,  $p_i, p_j$  if and only if there is no other vertex within the *lune* between them.

$$(p_i, p_j) \in \text{RNG}(P) \\ \iff |p_i - p_j| \leq \max_{k \neq i, j} (|p_i - p_k|, |p_j - p_k|)$$

Figure ?? shows a highlighted 2D example of a connections and precluded connection.

Call the relative neighborhood graph around a point to be  $\text{RNG}(P, p)$ . That is, the subset of the relative neighborhood graph that only has an edge that includes  $p$ .

When the points are drawn from a uniform random process in the unit cube, the maximum degree of the relative neighborhood graph is bounded by a constant.

## 2. Related Work

## 3. Motivation and Structure

For  $P = (p_0, p_1, \dots, p_{n-1})$ , a rough overview of the algorithm proceeds as follows:

- Create a grid, where each cell size of  $n^{-1/3}$
- Bin points into grid cells, allowing for multiple points in a bin
- For each point,  $p$ :
  - keep adding nearby points from adjacent grid cells and discard further away points whose connection to  $p$  would be precluded by the existence of the nearby point

- keep extending the radius of a bounding cube, at grid cell increments, until the collected nearby points have precluded connections from all other points outside bounding cube to  $p$
- once a bounding cube has been found that has precluded a connection to  $p$  from all other points outside, add nearby points outside the bounding cube that might preclude a connection of inside points to  $p$
- run a naive relative neighborhood graph,  $\text{RNG}(P, p)$ , on all collected nearby points to determine connections to  $p$

A coarse grained plane partition heuristic is used to exclude further away points from consideration.

Given points  $p$  and  $q$ , any point,  $w$ , that lies on the other side of the plane with normal  $\frac{(q-p)}{|q-p|}$  that passes through  $q$ , cannot connect to  $p$ , as  $q$  would be in any lune that  $p$  and  $w$  would create. Figure ?? gives a graphical representation of points excluded by this plane heuristic in two dimensions.

The plane partition heuristic is coarse, as the entire region that  $q$  excludes from  $p$  is more complicated (see Figure ??). While the plane partition heuristic is coarse, it's easy to calculate and provides a good enough way to exclude large portions of points so that it can be used as the foundation for an expected linear time algorithm.

To maintain a linear time algorithm, we need to avoid processing points individually and to implicitly exclude large collection of points that can never connect to the our anchor point  $p$ . To do this, we consider a current grid cell collection of integral radius  $R \in \mathbb{Z}$ , centered on the grid cell where anchor point  $p$  resides. If the tangent planes that passes through neighbor  $q$ , and in direction  $(q - p)$  within the grid cell collection, partition each of the six sides of our cube of radius  $R$ , we can be assured that any points outside of the grid cell collection will never be able to connect to  $p$ .

A grid cell collection of a radius  $R$  is called a *fence* of radius  $R \in \mathbb{Z}$ . When  $R$  is understood from context, this will be called just a *fence*. If the sides of the *fence* has been

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partitioned, with anchor point  $p$  and neighbors  $q$  within the fence, the fence will be called *secured*.

Points outside of the fence will never be able to connect to anchor point  $p$  but they still might be in the lune of some points within the fence. For this reason, when collecting points for the naive relative neighborhood graph calculation, some points outside of the fence will need to be added, as they can sabotage potential edges within the fence.

This will be done by considering a larger radius of the current fence to add points but the increase in  $R$  will shown to be bounded by a constant factor, allowing overall linear time to be maintained.

Before discussing the algorithm in detail, preliminary structural results will be discussed.

If  $w$  is on the other side of the plane that passes through  $q$  and with tangent in the direction of  $(q - p)$ , then  $q$  will always be in the lune created by  $p$  and  $w$  (see Figure ??):

**Lemma 1 (Plane Partition Heuristic).** *if  $p, q, w \in \mathbb{R}^3$  with:*

$$w \in H_{p,q}^+ = \{u : (q - p) \cdot (u - q) \geq 0\}$$

then

$$q \in \text{Lune}(p, w) = \{v : |v - p| \leq |p - w| \ \& \ |v - w| \leq |p - w|\}$$

PROOF. TK

The plane partition heuristic allows us to construct a convex hull from neighboring points that will bound the area that needs to be tested for relative neighborhood graph neighbors of  $p$ .

Only points within the convex hull have the potential to connect to  $p$ :

**Lemma 2 (Convex Hull Heuristic).** *Let  $p \in \mathbb{R}^3$ ,  $Q = (q_0, q_1, \dots, q_{m-1}) \subset P$  and  $C$  be the compact convex hull created from  $H_i^- = \{u \in \mathbb{R}^3 | (q_i - p) \cdot (u - q_i) < 0\}$ , then  $(w, p) \in \text{RNG}(P, p) \rightarrow w \in C$*

PROOF. By Lemma 1, all  $w$  outside of  $C$  will have some  $q_i \in Q$  in the  $\text{Lune}(w, p)$ , so cannot have an RNG edge to  $p$ .

Lemma 2 only guarantees that RNG connections to  $p$  fall within the hull,  $C$ . Not every point within  $C$  might connect to  $p$  and points outside of  $C$  might still influence connections to  $p$ . For example, Figure ?? shows a potential connection between two points within the convex hull but is thwarted by a point outside the convex hull that is within the lune created by the pair.

Points outside of the convex hull might influence connections to  $p$  but we can bound how far these points can be. The maximum distance to  $p$  and the convex hull bounds what points can influence connections to  $p$  to that distance:

**Lemma 3 (Convex Hull Extension Heuristic).** *Let  $p, C$  be the anchor point and convex hull as detailed above. Let  $R_{\max} = \max_{u \in C} |u - p|$  be the maximum distance of the*

*convex hull boundary to  $p$ . All points that could affect  $\text{RNG}(P, p)$  lie within a sphere of radius  $R_{\max}$ , centered at  $p$ .*

PROOF. The maximum lune radius for any neighbor to  $p$  is bounded above by  $R_{\max}$ . Any point,  $w$  such that  $|w - p| > D$ , will be further away from the maximum lune radius and so can't influence any connection within  $C$ .

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## 4. Algorithm

**Algorithm 1** Relative Neighborhood Graph for single point, Naive method

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1: procedure RELATIVENEIGHBORHOODGRAPHPOINT-
   NAIVE( $L, p$ )
2:   for  $q \in L / \{p\}$  do
3:     isConnected = False
4:     for  $u \in L / \{p, q\}$  do
5:       if  $\text{InLune}(p, q, u)$  then
6:         isConnected = False
7:       if isConnected = True then  $E = E \cup (p, q)$ 
8:   return  $E$ 

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**Algorithm 2** Relative Neighborhood Graph, Secure Posts on Increasing Fence (SPoIF)

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1: procedure SPoIF_RNG( $P$ )
2:    $E = \{\}$ 
3:    $N = |P|$ ,  $N_s = \lceil N^{\frac{1}{3}} \rceil$ ,  $s = \frac{1}{N_s}$ ,
4:    $B[0 : N_s][0 : N_s][0 : N_s] = \emptyset$ 
5:   for  $p \in P$  do
6:     append( $B[\lfloor N_s p_x \rfloor][\lfloor N_s p_y \rfloor][\lfloor N_s p_z \rfloor], p$ )
7:   for  $p \in P$  do
8:      $E = E \cup \text{SPoIF\_RNGp}(B, P, p)$ 
9:   return  $E$ 

```

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## 5. Conclusion

## 6. Bibliography

### A. Appendix.0

For completeness, this section discusses the elementary but tedious calculations for the plume shape and volume. As a reminder, our goal is to show that increasing the grid fence by a fixed size will almost surely encapsulate a compact convex hull made from an anchor point and its neighbors. The convex hull guarantees that the relative neighborhood graph edges of the anchor point to its neighbors need only consider candidate points within the convex hull (with some caveats about extending the grid fence to include saboteurs, detailed in section ??).

**Algorithm 3** Single Point Relative Neighborhood Graph, Secure Posts on Increasing Fence, single point

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1: procedure SPOIF_RNGP( $B, p$ )
2:   Initialize FencePostCluster to False,
3:   take  $2 \times 2$  clusters of fence posts for each face
4:    $R_f = 0, R_{\max} = 0$ 
5:   for ( $R_f \leq N_s$ ) & ( $False \in \text{FencePostCluster}$ ) do
6:     for  $q \in \text{GridPerimeterCells}(B, p, R_f)/\{p\}$  do
7:        $L = L \cup \{q\}$ 
8:        $R_{\max} = \text{maximum of } R_{\max} \text{ and grid cell}$ 
       distance that lune of  $(p, q)$  intersects
9:       for  $q \in \text{GridPerimeterCells}(p, R_f)/\{p\}$  do
10:        Define  $\rho(u) = \text{sgn}((q - p) \cdot (q - u))$ 
11:        for every cluster,  $FPC$  do
12:          if every  $v \in FPC, \rho(v) = -\rho(p)$  then
13:            Mark cluster in FencePostCluster as
            True
             $R_f ++$ 
14:   for  $R_e = [R_f + 1 : R_{\max}]$  do
15:     for  $q \in \text{GridPerimeterCells}(p, R_f)/\{p\}$  do
16:        $L = L \cup \{q\}$ 
17:   return RNGPointNaive( $P, B, p, L$ )
    
```

The logic chain is roughly:

- For an anchor point,  $p$ , and a grid fence side, there will be a region that a neighbor point can fall in that will partition the fence side from the point, a process we call *securing the fence*, where a secured fence necessarily implies a compact convex hull made from the points on the interior
- The volume that secures a fence side will turn out to be four sphere slices joined together, which we will call a *plume*, where a sphere slice is the intersection of two perpendicular half planes and a sphere
- A lower bound estimate on the plume volume is a constant proportion of overall grid fence volume
- A non-trivial constant plume volume makes the question of how many neighbors to be expected to secure the overall six sided fence into a coupon collector-like problem, giving us an expected finite grid fence size
- Finally, approximations on the four slices bundled together allow us to give coarse values for grid fence sizes

It should be stressed that the discussion and estimates are meant primarily to show existence of a finite grid fence size. We focus on securing a fence side by waiting for some single neighbor point to secure it. The Shrinking Posts on Increasing Fence (SPOIF) algorithm in this paper has a smaller expected grid fence size before securitization as points can secure parts of the fence through clusters of posts, increasing the admissible region on each face side that can be secured.

The calculations done in this appendix are meant to show that we only need to consider a finite grid fence size by providing correct, if horrible, bounds. In this paper, we are satisfied with providing empherical results for grid sizes that SPOIF uses without going into more rigorous analysis.

First consider a fixed point  $q$  and a fixed but moveable point  $\eta$ .

We think of  $\eta$  as a point on a plane with normal  $\eta/|\eta|$  that intersects  $q$  in the slice of  $z = 0$  to  $z = 1$ .

As we vary  $\eta$  we want to know what the shape is traced out.

The claim is that this is the surface of a sphere whose center is at  $q/2$  of radius  $q/2$ .

To see this, consider  $\eta = q/2 + (|q|/2)v$ , with  $|v| = 1$ .

Calculate dot product of the line from  $q$  to  $\eta$  with the line from the origin to  $\eta$ :

$$\begin{aligned}
 \eta &= q/2 + (|q|/2)v \\
 (q/2 + (|q|/2)v) \cdot (q/2 + (|q|/2)v - q) \\
 &= (q/2 + (|q|/2)v) \cdot (-q/2 + (|q|/2)v) \\
 &= -(1/4)|q|^2 + (|q|/2)^2|v|^2 \\
 &= 0
 \end{aligned}$$

Proving the claim.

We want to find the the minimum volume of the plume for some point inside of a fence of radius  $s$  (side length  $2s$ ) nested inside of a larger fence of radius  $s(2k+1)$  (side length  $2s(2k+1)$ ).

First consider a sphere of radius  $R$  centered at  $(-D, -D, -D)$  intersected with two half planes,  $H_{x=0}^+ = \{u : u_x \geq 0\}$ ,  $H_{y=0}^+ = \{u : u_y \geq 0\}$ . For suitable  $R$  and  $D$ , we want to estimate the volume.

If we take two tetrahedrons completely enclosed by the slice, the maximum diagonal of the slice is  $(R - \sqrt{2}D)$ , with maximum straight side length (in the  $x$ -axis line and  $y$ -axis line) to also be  $(R - \sqrt{2}D)$  and slice half height to also be  $(R - \sqrt{2}D)$ .

The volume of the tetrahedron is then:

$$(1/6)(R - \sqrt{2}D)^2 \sqrt{R^2 - 2D^2}$$

The volume of the two tetrahedrons stacked on top of each other:

$$(1/3)(R - \sqrt{2}D)^2 \sqrt{R^2 - 2D^2}$$

More succinctly:

$$V_{\text{slice}} > \frac{1}{3}(R - \sqrt{2}D)^2 \sqrt{R^2 - 2D^2}$$

We will get concrete estimates below but as a quick confirmation, we can check to make sure that this slice is a constant proportion of the fence it sits in.

We know that  $R \propto (ks)$  and  $D \propto (ks)$ , for some  $k \in \mathbb{Z}$  and grid cell size  $s \in \mathbb{R}$ . Call  $R = \alpha(ks)$  and  $D = \beta(ks)$ .

The plume will be four slices bundled together sitting inside of a  $\gamma^3(ks)^3$  volume. Take  $\alpha, \beta \in \mathbb{R}$  to be the smallest of the four slices:

$$\begin{aligned}
 & \frac{(4/3)(\alpha(k s) - \sqrt{2\beta(k s)})^2 \sqrt{\alpha^2(k s)^2 - 2\beta^2(k s)^2}}{\gamma^3(k s)^3} \\
 &= \frac{(4/3)(k s)^3 (\alpha - \sqrt{2\beta})^2 \sqrt{\alpha^2 - 2\beta^2}}{\gamma^3(k s)^3} \\
 &= \frac{(4/3)(\alpha - \sqrt{2\beta})^2 \sqrt{\alpha^2 - 2\beta^2}}{\gamma^3}
 \end{aligned}$$

as desired.

When choosing random points inside of the larger fence cube, we can think of the plume as the probability a single point secures one side of the fence. It is equivalent to talk about securing one side of the fence face with securing four posts on each of corners of the fence face. As a reminder, the Shrinking Posts on Increasing Fence (SPoIF) uses nine posts per face, equally distributed along the face square. Since there are more opportunities to secure clusters of posts on a fence face, in addition to still retaining the features of the plume region, SPoIF will be able to secure the fence cube more quickly. We only consider the four post fence here to help simplify the calculation even though it inflates the expected number of points we need to see to secure the grid fence.

Take the minimum plume value volume of the six directions and call it  $V_{\text{plume}}$ . The probability that a point lands in one of the plumes is bounded below by  $V_{\text{plume}}/V_{\text{fence}}$ . Of the points that land in one of the plumes, the time we need to take to secure each face of the fence is now a coupon collector-like problem with six coupons.

We can now plug in some real numbers to get coarse bounds on plume volumes, probabilities of landing in the critical region and expected grid sizes.

As a reminder, we are considering a center grid cell, of side length  $s$  where the anchor point,  $p$ , resides, all of which is inside a larger grid fence of side length  $2s(k+1)$ , for some fixed  $k \geq 0, k \in \mathbb{Z}$  as yet to be determined.

The smallest size a slice can be is if the anchor point is on the corner of the interior grid cell and to choose the grid fence corner nearest to it. For example,  $p = (s/2, s/2, s/2)$ ,  $q = (s(k+1/2), s(k+1/2), s(k+1/2))$ .

In such a case, the center point of the sphere is  $(p+q)/2 = (s(k+1)/2, s(k+1)/2, s(k+1)/2)$ , with radius  $R = |(p+q)/2 - p| = |(sk/2, sk/2, sk/2)| = (\sqrt{3}/2)sk$ . Here,  $D = s(k+1)/2 - s/2 = sk/2$  with  $V_{\text{fence}} = (s(2k+1))^3$ .

This yields:

$$\begin{aligned}
 \frac{V_{\text{plume}}}{V_{\text{fence}}} &> \frac{\frac{4}{3}(\frac{\sqrt{3}}{2}sk - \frac{\sqrt{2}}{2}sk)^2 (\frac{3}{4}s^2k^2 - \frac{1}{2}s^2k^2)^{\frac{1}{2}}}{s^3(2k+1)^3} \\
 &= \frac{s^3k^3(\sqrt{3}-\sqrt{2})^2}{3\sqrt{2}s^3(2k+1)^3} \\
 &= \frac{(\sqrt{3}-\sqrt{2})^2}{3\sqrt{2}(2+\frac{1}{k})^3} \\
 &> \frac{(\sqrt{3}-\sqrt{2})^2}{81\sqrt{2}} \quad (k \geq 1) \\
 &> 1/1198
 \end{aligned}$$

For points that land within one of the six plume region, we would expect this to be a six coupon collector problem with, at worst, equal probability. The coupon collector problem grows roughly as  $n \log(n)$  which, in our case of six sides of the fence to secure, gives us an expected 11 or more points before we secure the fence.

Each point has roughly a  $6/1198$  chance of landing in one of the six plumes, so we would expect roughly  $11/(6/1198) = 2196.3 \dots$  number of points within the grid fence to appear before securing the fence.

Each grid cell has roughly one point, so we would expect the number grid cells on a side to be

$$\begin{aligned}
 & (11 \cdot 1198/6)^{\frac{1}{3}} \\
 &= 12.9 \dots \sim 13
 \end{aligned}$$

That is, we would expect  $k \sim 6$ , iterating roughly six to seven steps of increasing the grid fence size until securing the fence in this way.

## B. Appendix.1

... Appendix sections are coded under \appendix.

## References

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